

DEMO · Flight Data Recorder Dataframe Document

Dataframe: DEMO_256WPS
Revision: A Issue date: 2026-09-11
Recording format: ARINC 717, 256 WPS, 4 subframes of 1 second
Sync words: 583, 1464, 2631, 3512
Bit numbering: 12 (MSB) .. 1 (LSB). Engineering value = decimal × resolution + offset.

Parameter	Description	Type	Freq (Hz)	Word	Subframe	MSB	LSB	Units	Resolution	Offset	Min	Max	1 =	0 =	Remarks
PITCH ATT #1	Pitch attitude, captain side	Signed Analog	2	4, 132	all	12	3	deg	0.176	0	-90	90			
ROLL ATT	Roll attitude	Signed Analog	1	5	all	12	3	deg	0.176	0	-180	180			
HDG MAG	Magnetic heading	Unsigned Analog	0.5	6	1,3	12	1	deg	0.0879	0	0	360			
IAS	Indicated airspeed	Unsigned Analog	1	7	all	12	1	kt	0.25	0	0	450			
PRESS ALT	Pressure altitude, coarse+fine multi-segment	Signed Analog	1	154, 153	all	9/12	1/1	ft	0.25	0	-2000	60000			
FLAP POS	Flap surface position	Unsigned Analog	0.5	10	2,4	12	1	deg	0.0062	-1.6	-2	45			
AOA	Angle of attack	Unsigned Analog	0.5	11	1,3	12	1	deg	0.0187	-10.8	-11	41			
AILERON POS	Aileron position (negative resolution)	Unsigned Analog	1	12	all	12	1	deg	-0.0153	31.2	-31.5	31.5			

DEMO_256WPS (continued)

Parameter	Description	Type	Freq (Hz)	Word	Subframe	MSB	LSB	Units	Resolution	Offset	Min	Max	1 =	0 =	Remarks
LDG GEAR DOWN	Landing gear lever position	Discrete	1	13	all	1	1		1	0			DOWN	UP	
GPWS WARN	Ground proximity warning (active low)	Discrete	1	13	all	2	2		1	0			NORMAL	WARNING	Active low: raw 0 means warning active
SEL CRS	Selected course (BCD, 3 digits)	BCD	0.25	14	2	12	1	deg	1	0	0	359			
SPARE 15	Unassigned source type (decodes as UNSUPPORTED TYPE)	Special	0.25	15	1	12	1		1	0					